

ECG Simulator - Comprehensive Documentation

Executive Summary

The ECG Simulator is a Python-based biomedical simulation framework modeling cardiac electrophysiology and generating realistic 12-lead electrocardiograms.

Two computational engines are implemented:

- **Phase 1/2:** Discrete conduction graph with BFS propagation (microseconds per step, extremely fast)
- **Phase 3:** FitzHugh-Nagumo (FHN) 1D cable model with emergent refractory (2.2x real-time, physiologically faithful)

Mathematical Models

Cardiac Action Potential

Every myocyte exhibits an action potential (AP) with five phases:

1. Phase 0: Rapid depolarization (-90 mV to +30 mV) via inward Na⁺ current
2. Phase 1: Early repolarization via outward K⁺ current
3. Phase 2: Plateau (100-300 ms): inward Ca²⁺ balances outward K⁺
4. Phase 3: Late repolarization via dominant outward K⁺
5. Phase 4: Resting diastolic potential at -90 mV

Conduction and Refractoriness

Electrical activation propagates via gap junctions between cells at characteristic velocities:

- Atrial myocardium: 0.5-1.0 m/s
- His-Purkinje system: 2-4 m/s (fastest)
- Ventricular myocardium: 0.3-0.5 m/s
- AV node: 0.02-0.05 m/s (slowest)

During and after the AP, the cell is refractory (cannot re-excite). The absolute refractory period (during phases 0-2) prevents premature activation. Refractory period duration correlates with AP duration (APD).

ECG and Dipole Theory

The 12-lead ECG measures the projection of the cardiac electrical dipole onto fixed spatial lead vectors. Each anatomical region (SA node, RA, LA, AV node, bundle branches, ventricles) contributes a time-localized dipole:

$$\mathbf{p}_{\text{node}}(t) = \mathbf{d}_{\text{node}} [g_{\text{depol}}(t - t_{\text{act}}) + A_{\text{repol}} g_{\text{repol}}(t - t_{\text{act}})]$$

where:

- $\mathbf{d}_{\text{node}}$ = dipole vector (anatomy-specific)
- t_{act} = activation time
- $g_{\text{depol}}, g_{\text{repol}}$ = Gaussian functions (QRS and T-wave)
- Depolarization duration $\sigma_d \approx 12.7$ ms
- Repolarization peaks at ≈ 280 ms post-activation

The total ECG voltage in lead k is the sum over all nodes:

$$\text{ECG}_k(t) = \sum_{\text{nodes}} \mathbf{L}_k \cdot \mathbf{p}_{\text{node}}(t)$$

Phase 3: FitzHugh-Nagumo Cable Model

Mathematical Formulation

The 1D monodomain FHN equations are:

$$\begin{aligned} \frac{\partial v}{\partial t} &= \frac{D}{\tau} \frac{\partial^2 v}{\partial x^2} + \frac{1}{\tau} \left(v - \frac{v^3}{3} - w + I_{\text{ext}} \right) \\ \frac{\partial w}{\partial t} &= \frac{\varepsilon}{\tau} (v + a - b \cdot w) \end{aligned}$$

where:

- $v(x, t)$ = transmembrane voltage (normalized: $-1.2 \leq v \leq 2$)
- $w(x, t)$ = slow recovery variable
- D = diffusion coefficient (cell-to-cell coupling)
- τ = time scale (15 ms in Phase 3-D calibration)
- ε = recovery rate (0.08)
- $a = 0.7, b = 0.8$ = FHN model parameters
- I_{ext} = external stimulus

Fast-Slow Dynamics

The key insight: v evolves on the fast timescale $\sim \tau$, while w evolves on the slow timescale $\sim \tau/\varepsilon$.

- **Depolarization:** v rapidly increases as depolarizing current dominates
- **Plateau:** v remains high; w slowly increases, gradually opposing depolarization
- **Repolarization:** When w is large enough, it repolarizes v
- **Refractoriness:** w remains large, suppressing v . As w decays, v can respond again

The AP duration (APD) is determined by recovery dynamics:

$$\text{APD} \sim \frac{\tau}{\varepsilon \cdot b} = \frac{15 \text{ ms}}{0.08 \times 0.8} \approx 234 \text{ ms}$$

This natural recovery time generates multi-beat rhythms without explicit programming.

Cable Structure

Atrial cable: 8 cells

- Cell 0: SA node (pacemaker, point source with $D = 0$)
- Cells 1-7: RA and LA myocardium
- Diffusion: $D_{\text{atr}} = 2.88$

Ventricular cable: 10 cells

- Cell 0: His bundle (point source, $D = 0$)
- Cells 1-3: Left bundle branch (fast, $D = 72$)
- Cells 4-9: LV myocardium (slower, $D = 3$)

Numerical Integration

Discretized using finite differences with explicit Euler integration. The CFL (Courant-Friedrichs-Lewy) stability condition requires:

$$D \cdot \frac{\Delta t}{\tau \Delta x^2} \leq 0.5$$

Current values with $\Delta t = 0.1$ ms, $\tau = 15$ ms, $\Delta x = 1$:

$$\text{CFL}_{\text{ven}} = 72 \times \frac{0.0001}{0.015 \times 1} = 0.48 < 0.5$$

Python Implementation

Core Modules

File: `cardiac_sim/core/tissue/cable_1d.py` (500 lines)

- Class: `ConductionCable1D(params)`
- State: atrial voltage/recovery arrays, ventricular arrays
- Key methods:
 - `new_beat(sa_fire_time, params)`: Reset for new beat
 - `advance(dt)`: Integrate by dt seconds
 - `_step_atrial(dt)`, `_step_ventricular(dt)`: FHN Euler steps
- Integration: `_fhn_cable_step(V, W, D, I, dt, ...)` with optional Numba JIT

File: `cardiac_sim/simulation/cable_engine.py` (450 lines)

- Class: `ConductionCableEngine(SimulationEngine)`
- Manages beat lifecycle, node activation tracking, ECG computation
- Key methods:
 - `initialize(params)`: Create cable and node dictionary
 - `step(dt)`: Main loop - advance cable, collect activations, compute ECG
 - `_fire_sa()`: Initiate new beat
 - `_compute_ecg(t)`: Sum Gaussian dipoles from active beats

GUI Components

File: `cardiac_sim/gui/main_window.py` (250 lines)

- Main application window with three regions:
 - Toolbar: Run/Pause, Reset, Engine selector, HR display
 - Left sidebar: Parameter panel, Pathology panel
 - Central area: 12-lead ECG real-time display (pyqtgraph)

File: `cardiac_sim/core/parameter_model.py` (200 lines)

- Complete parameter hierarchy as dataclasses
- Validation ranges
- Default physiological values

Parameters and Ranges

All major parameters with physiological meaning:

SA Node:

- Cycle Length (ms): 300-2000 [default: 857 → 70 bpm]
- Ectopic Trigger Rate (bpm): 0-200

AV Node:

- AV Delay (ms): 50-300 [default: 100]
- AV Conductance: 0-1 [0=block, 1=normal]
- AV Refractory Period (ms): 200-400

Ventricular:

- RBB/LBB Conductance: 0-1 [blocks implemented]
- Septal Conductance: 0-1
- Ventricular Refractory (ms): 200-350

Phase 3 Model:

- Cable Time Scale τ (ms): 5-30 [default: 15]
- Atrial Diffusion: 0.5-10 [default: 2.88]
- Ventricular Diffusion (LBB): 50-150 [default: 72]
- Ventricular Diffusion (LV): 1-20 [default: 3]

Pathological ECG Patterns

Available pathologies reproducible via the simulator:

AV Conduction Blocks

- **1° AV Block:** AV delay increased. PR $\geq$ 200 ms, all P followed by QRS. Reproduction: Parameters → AV Node → AV Delay = 150 ms
- **3° AV Block:** AV conductance = 0. Atrial and ventricular rhythms completely independent. Reproduction: Parameters → AV Node → AV Conductance = 0

Bundle Branch Blocks

- **RBBB:** RBB conductance = 0. RV activation delayed. QRS ≥ 120 ms, terminal R' in V1. Reproduction: Parameters $\rightarrow$ Ventricular $\rightarrow$ RBB Conductance = 0
- **LBBB:** LBB conductance = 0. LV activation delayed. QRS ≥ 120 ms, broad R in lateral leads. Reproduction: Parameters $\rightarrow$ Ventricular $\rightarrow$ LBB Conductance = 0

Atrial Arrhythmias

- **Atrial Fibrillation:** No P waves, erratic baseline, irregularly irregular QRS rhythm
- **Sinus Bradycardia:** All intervals normal, just slower rate (e.g., 50 bpm)
- **Sinus Tachycardia:** All intervals normal, faster rate (e.g., 140 bpm)

Ventricular Arrhythmias

- **PVC:** Wide abnormal QRS; opposite T wave; compensatory pause
- **Ventricular Fibrillation:** Chaotic baseline, no organized QRS

Phase 3-D Validation Results

Normal sinus rhythm at 70 bpm:

Interval	Simulator	Clinical Target
P wave	64 ms	40-100 ms
PR interval	196 ms	120-200 ms
QRS duration	70 ms	≤ 120 ms
QT interval	360 ms	300-440 ms
HR	70 bpm	60-100 bpm

ECG morphology:

- QRS peak-to-trough: 1.3 mV (clinical: 0.5-2.5 mV)
- T wave: 0.6 mV
- QRS/T ratio: 2.2 (clinical normal: 1-2.5)
- 6 consecutive beats at 70 bpm over 5 seconds

Installation and Usage

Prerequisites: Python 3.12+, PyQt6, NumPy, pyqtgraph

```
git clone <repo>
cd ECG_simulator
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
python main.py
```

File Organization

Total: $\approx$ 3500 lines of code across 19 Python modules:

cardiac_sim/core/tissue/cable_1d.py	: 500 lines (FHN cable)
cardiac_sim/simulation/cable_engine.py	: 450 lines (Phase 3 engine)
cardiac_sim/gui/main_window.py	: 250 lines (GUI)
cardiac_sim/core/parameter_model.py	: 200 lines (Parameters)
cardiac_sim/simulation/graph_engine.py	: 400 lines (Phase 1/2 engine)
cardiac_sim/core/conduction/graph.py	: 150 lines (Graph structure)
cardiac_sim/gui/ecg_display.py	: 200 lines (ECG plotter)
cardiac_sim/core/cell_models/fitzhugh_nagumo.py	: 100 lines
cardiac_sim/core/ecg/lead_field.py	: 120 lines
cardiac_sim/plugins/*.py	: 400 lines (Pathologies)
[Others]	: 635 lines

Key References

FitzHugh, R. (1961). Impulses and physiological states in theoretical models of nerve membrane. *Biophys. J.*, 1(6), 445-466.

Keener, J. P., & Sneyd, J. (2009). *Mathematical Physiology: I. Cellular Physiology* (2nd ed.). Springer.

Roth, B. J. (1997). Electrical conductivity values used with the bidomain model of cardiac tissue. *IEEE Trans. Biomed. Eng.*, 44(4), 326-338.